The Importance of the Nitrous Oxide Pathway to NO_x in Lean-Premixed Combustion

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This study addresses the importance of the different chemical pathways responsible for NO_x formation in lean-premixed combustion, and especially the role of the nitrous oxide pathway relative to the traditional Zeldovich pathway. NO_x formation is modeled and computed over a range of operating conditions for the lean-premixed primary zone of gas turbine engine combustors. The primary zone, of uniform fuel–air ratio, is modeled as a micromixed well-stirred reactor, representing the flame zone, followed by a series of plug flow reactors, representing the postflame zone. The fuel is methane. The fuel–air equivalence ratio is varied from 0.5 to 0.7. The chemical reactor model permits study of the three pathways by which NO_x forms, which are the Zeldovich, nitrous oxide, and prompt pathways. Modeling is also performed for the well-stirred reactor alone. Three recently published, complete chemical kinetic mechanisms for the C1–C2 hydrocarbon oxidation and the NO_x formation are applied and compared. Verification of the model is based on the comparison of its NO_x output to experimental results published for atmospheric pressure jet-stirred reactors and for a 10 atm. porous-plate burner. Good agreement between the modeled results and the measurements is obtained for most of the jet-stirred reactor operating range. For the porous-plate burner, the model shows agreement to the NO_x measurements within a factor of two, with close agreement occurring at the leanest and coolest cases examined. For lean-premixed combustion at gas turbine engine conditions, the nitrous oxide pathway is found to be important, though the Zeldovich pathway cannot be neglected. The prompt pathway, however, contributes small-to-negligible NO_x. Whenever the NO_x emission is in the 15 to 30 ppmv (15 percent O_2, dry) range, the nitrous oxide pathway is predicted to contribute 40 to 45 percent of the NO_x for high-pressure engines (30 atm), and 20 to 35 percent of the NO_x for intermediate pressure engines (10 atm). For conditions producing NO_x of less than 10 ppmv (15 percent O_2, dry), the nitrous oxide contribution increases steeply and approaches 100 percent. For lean-premixed combustion in the atmospheric pressure jet-stirred reactors, different behavior is found. All three pathways contribute; none can be dismissed. No universal behavior is found for the pressure dependence of the NO_x. It does appear, however, that lean-premixed combustors operated in the vicinity of 10 atm have a relatively weak pressure dependence, whereas combustors operated in the vicinity of 30 atm have an approximately square root pressure dependence of the NO_x.

Introduction

During the 1992 Gas Turbine and Aeroengine Congress, it was stated in more than one presentation that the nitrous oxide pathway is responsible for much of the NO_x formed in lean-premixed gas turbine engine combustors fired on natural gas. This possibility is examined in this study by using a simple chemical reactor model, and the recently published kinetic mechanisms of hydrocarbon oxidation and NO_x formation of Drake and Blint [1], Miller and Bowman [2], and Michaud et al. [3].

The benefits of this type of study are as follows:

1 It permits the determination of combustion conditions for which the nitrous oxide pathway should be considered, and for which the Zeldovich pathway alone is insufficient to explain the NO_x. It indicates the importance that should be given to including the nitrous oxide chemistry in fluid codes for gas turbine combustors.

2 It permits the NO_x trends for lean-premixed combustion to be studied, for example, the pressure dependence of the NO_x.

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3 It provides insight into combustor arrangements that have the potential to minimize the formation of NO_x.

This paper is arranged so that the chemical reactor model and chemical kinetic mechanisms are summarized first. This is followed by comparisons of the model predictions to the measurements of NO_x obtained for well-defined laboratory burners. This provides for the verification of the model. This is followed by the application of the model to the prediction of NO_x for lean-premixed combustion conditions.

The focus of this study is on the chemical kinetics of NO_x formation. Chemical pathways most likely to form the NO_x in lean-premixed combustors are determined. Results obtained with different chemical mechanisms are compared. The fluid mechanics of NO_x formation is not modeled in this study. That is left to other studies. However, inference can be drawn from the results obtained herein regarding the role fluid mechanics might play in minimizing the NO_x formation in lean-premixed combustors.

Chemical Reactor Model

The baseline chemical reactor model employed consists of a micromixed well-stirred reactor (WSR), followed immediately by a series of plug-flow reactors (PFR). Usually, the WSR is assigned the smallest volume that will permit reaction for a given mass flow rate of reactants. This is termed "incipient blowout." It is defined to be 101 percent of the volume for blowout of the WSR.

The WSR provides for the initiation and stabilization of the combustion process, and represents the flame zone. The PFR provides for the continuation of the combustion process, and given adequate mean residence time, permits the combustion process to approach completion. It represents the post flame zone. Together, the WSR and PFR, of uniform fuel–air equivalence ratio, are taken to simulate the complete lean-premixed combustion process.

A variation of the baseline model is used to assess the sensitivity of the NO_x to the fraction of the combustor volume assumed to be the WSR. In this variation, the WSR component is increased progressively until it comprises the complete combustor, at which point the PFR component is eliminated. In an actual combustor, this variation would require the inlet streams to penetrate increasingly deeper into the combustor, and thus, produce an increasingly larger region that is intensely mixed and is high in free radical concentrations. The effect of this, as shown later, is to increase the NO_x.

Another variation of the model involves removal of the WSR. Combustion stabilization is then accomplished by including one or more recycle zones, by which hot, burned gases are brought forward to the initial plug-flow reactor stage. In a previous study of lean-premixed combustion for gas turbine engines [4], this variation was found to cause little change in the NO_x compared to that predicted by the baseline model, and thus, is not reconsidered herein.

In order to validate the baseline model, and the WSR model alone, test cases were computed and compared to datasets for laboratory combustors. These datasets include atmospheric pressure jet-stirred reactor measurements from two laboratories, and porous-plate burner data for 10 atm. Following this phase of the study, the baseline model was applied to a range of lean-premixed gas turbine combustor operating conditions in order to determine the following:

1 The effect of varying the volume of the flame zone (i.e., the WSR) relative to the volume of the postflame zone (i.e., the PFR).
2 The effects of varying the combustor pressure (1, 10, and 30 atm) and inlet temperature (300, 600, and 800 K).

The NO_x Chemical Pathways

Zeldovich. The extended Zeldovich pathway is given below by reactions (1)–(3). Thermal NO_x refers to NO_x formed from the extended Zeldovich pathway when the O atom, H atom, OH radical, and O_2 are at chemical equilibrium concentrations. When these species have nonequilibrium concentrations, the NO_x formed is referred to as Zeldovich NO_x. Initiation occurs by the reverse rates of reactions (1) and (1*a*).

$$N + NO = N_2 + O \quad (1)$$

$$N + NO = N_2 + O(^1D) \quad (1a)$$

$$N + O_2 = NO + O \quad (2)$$

$$N + O_2 = NO + O(^1D) \quad (2a)$$

$$N + OH = NO + H \quad (3)$$

Note: $O(^1D)$ denotes the electronically excited singlet O atom; O denotes the ground state triplet O atom. Reactions (1*a*) and (2*a*) pertain only to the Michaud et al. kinetic mechanism [3].

Nitrous Oxide. The nitrous oxide pathway is given below by reactions (4)–(11). The mechanism is initiated by the reverse rates of reactions (4) and (4*a*), and the N_2O formed is then oxidized to NO mainly by reaction (5), but also forms by reactions (8), (10), and (11). The N_2O is reduced to N_2 by reactions (4), (4*a*), (6), (7), and (9).

$$N_2O + M = N_2 + O + M \quad (4)$$

$$N_2O + M = N_2 + O(^1D) + M \quad (4a)$$

$$N_2O + O = NO + NO \quad (5)$$

$$N_2O + O = N_2 + O_2 \quad (6)$$

$$N_2O + H = N_2 + OH \quad (7)$$

$$N_2O + H = NO + NH \quad (8)$$

$$N_2O + OH = N_2 + HO_2 \quad (9)$$

$$N_2O + CO = NO + NCO \quad (10)$$

$$N_2O + M = N + NO + M \quad (11)$$

Note: Reactions (4*a*) and (11) pertain only to the Michaud et al. kinetic mechanism [3].

Prompt. For the prompt mechanism, only the initiating reaction is listed. This is reaction (12) below. The balance of the prompt NO_x mechanism involves the oxidation of the HCN molecules and the N atoms, and the chemical recycle of NO to HCN by reaction with CH_x species. Under lean-premixed conditions, the HCN oxidizes to NO by the reaction sequence HCN => CN => NCO => NO. The N atom oxidizes mainly by reaction (2).

$$N_2 + CH = HCN + N \quad (12)$$

Nitrogen Dioxide. The NO formed by the three chemical pathways just described can oxidize to NO_2, mainly by reaction (13) listed below. Because of this, the mechanisms and results of this study are normally expressed as NO_x (the sum of NO and NO_2), rather than as NO.

$$NO + HO_2 = NO_2 + OH \quad (13)$$

Isolation of Pathways. In order to consider the NO_x contributed individually by these pathways, the following procedure can be used: (1) The Zeldovich NO_x is computed first by including only reactions (1–3) in the model. (2) This is followed by adding reactions (4–11) in order to give the sum of the Zeldovich NO_x and the NO_x contributed by the nitrous oxide pathway. (3) The concluding step involves the addition of the prompt NO_x pathway (i.e., reaction (12) and

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all of the additional reactions involving the nitrogen chemistry), to give the total NO_x. The individual contributions are then determined by difference.

A similar procedure, which gives results identical to those obtained by the long procedure just described (within 4 percent), regardless of the order of application of the following steps, is used in this study. Starting from the full mechanism, the steps are as follows:

1 Prompt NO_x is eliminated by removing reaction (12). This reaction is the primary source of the N atoms and cyano molecules formed by the attack of hydrocarbons on N_2. (Other, less important reactions between hydrocarbons and N_2 are also eliminated, even though they do not contribute significantly.)
2 Zeldovich NO_x is eliminated by removing the reverse rates of reactions (1) and (1*a*). These are the only reactions that produce N atoms and NO by attack of O atoms on N_2.
3 The contribution of the nitrous oxide pathway to NO_x is eliminated by removing the reverse rates of reactions (4) and (4*a*). These are the only reactions that contribute significantly to the formation of N_2O from N_2.

The Chemical Kinetic Mechanisms

Three complete chemical kinetic mechanisms of hydrocarbon oxidation and NO_x chemistry are used in the present study. These three mechanisms are those of Drake and Blint [1], Miller and Bowman [2], and Michaud et al. [3].

The kinetic mechanism of Drake and Blint, hereafter referred to as the DB mechanism, has been used without deletions. Reaction (9), with rate data as estimated in [2], has been added to the DB mechanism; and the pressure sensitivities of reaction (4) and the CH_4 dissociation reaction have been extended from 20 to 35 atm. The rate data selected for reaction (12) are from [5], since these have been found to give the best overall agreement between the model results and the jet-stirred reactor and porous-plate datasets considered. The DB kinetic mechanism, which was developed from verification of laminar, premixed flame data taken at the burner surface, shows an enhanced importance of the prompt NO_x pathway [1].

The kinetic mechanism of Miller and Bowman, hereafter referred to as the MB mechanism, has been used without deletions. The dissociation reactions for C_2H_6, C_2H_5, C_2H_3, CH_4, and reaction (14) listed below, have been corrected for elevated pressure. The MB kinetic mechanism is the successor to the Glarborg et al. mechanism [5], which was based on jet-stirred reactor data.

$$CH_2CO + M = CH_2 + CO + M \qquad (14)$$

The kinetic mechanism of Michaud et al., hereafter referred to as the MWF mechanism, has been used without deletions. The dissociation reactions for C_2H_6, C_2H_5, C_2H_3, CH_4, reaction (14), as well as reactions (15) and (16) listed below, have been corrected for elevated pressure. The MWF kinetic mechanism was developed using porous-plate burner data and includes the addition of the singlet O atom chemistry to the Zeldovich and nitrous oxide pathways.

$$OH + NO_2 = HOONO \qquad (15)$$

$$N_2O + H = HNNO \qquad (16)$$

These three kinetic mechanisms have been selected for this study, since they represent recent developments in NO_x chemistry for combustion sources. The MB mechanism has been used in several studies, the DB mechanism provides enhanced prompt NO_x rates, and the MWF mechanism provides alternative possibilities regarding the O atom attack on N_2 and the formation of N_2O. These mechanisms cover a useful range of chemical kinetic possibilities for NO_x formation in gas-fired, high-pressure combustors.

Laboratory Reactor NO_x Measurements

In the next two sections of this paper, results for NO_x are shown and discussed for two distinctly different types of laboratory combustion reactors. Results obtained by the model applied to the conditions of these reactors are examined. The two types of reactor are jet-stirred reactors and a porous-plate burner.

Jet-stirred reactors, when operated at short residence times (of a few milliseconds) can approach the well-stirred reactor (WSR) condition. At this condition, the rate of mixing is fast compared to the rate of chemistry for the process being studied, which in this case is the formation of NO_x. The condition is characterized as zero dimensional. Results for NO_x are shown in the next section, which appear to satisfy the WSR condition. Results are also shown for jet-stirred reactors operated so that the mixing is not fast compared to the chemistry. In this situation, the finite mixing rate, together with wall heat loss, promotes gradients and structure within the reactor and influences the NO_x formation. Increases in NO_x because of these influences are shown. However, decreases in NO_x because of a finite mixing rate can also occur, and are shown.

The porous-plate burner, on the other hand, is essentially a one-dimensional plug-flow reactor (PFR). Although the initial region of burner requires backmixing and back-diffusion for stabilization of the flame (the model uses a WSR zone), under lean-premixed conditions only a relatively small amount of the NO_x emitted from the burner is formed in this region. This region generally has a short mean residence time compared to that of the overall burner. Thus, the NO_x is formed predominantly in the bulk of the burner, which is well modeled as a PFR.

Jet-Stirred Reactor NO_x

Jet-stirred reactors permit NO_x to be examined under conditions of intense recirculation of the burning gases into the fresh reactants. Furthermore, they provide databases for the development of NO_x mechanisms, e.g., see [5].

Published results on NO_x formation for lean, premixed, methane-fired jet-stirred reactors are reviewed and compared to the model predictions. The model predictions are based on the measured temperatures, pressures, mean residence times, and inlet reactant concentrations. The NO_x yields are in the low (< 10 ppmv) and modest (< 100 ppmv) ranges. The data are based on the actual O_2 (dry) concentrations. The reactor pressure is ambient, which is either 1.0 or 0.92 atm (depending on the elevation of the laboratory site). Three types of results are shown, including: (1) NO_x trends, (2) comparison to the model, and (3) NO_x pathways.

NO_x Trends

Trends in the rate of formation of NO_x are shown in Fig. 1. This rate is based on the NO_x measured for the reactor divided by the mean residence time of the reactor. These results are derived from [6]. The measurements were obtained at a pressure of $P = 0.92$ atm, a fuel–air equivalence ratio of $\phi = 0.75$, an inlet temperature of the reactants of $T_{in} = 295$ K, and an inlet air humidity ratio of 8×10^{-4} g H_2O/g dry air. The mean residence time was varied over a wide range, from $\tau = 6$ to 40 ms. The reactor temperature depended mainly on the mean residence time. That is, with increasing mean residence time, the reactor experienced increasing heat loss, which led to a decrease in temperature.

The main trends evident from Fig. 1 are as follows:

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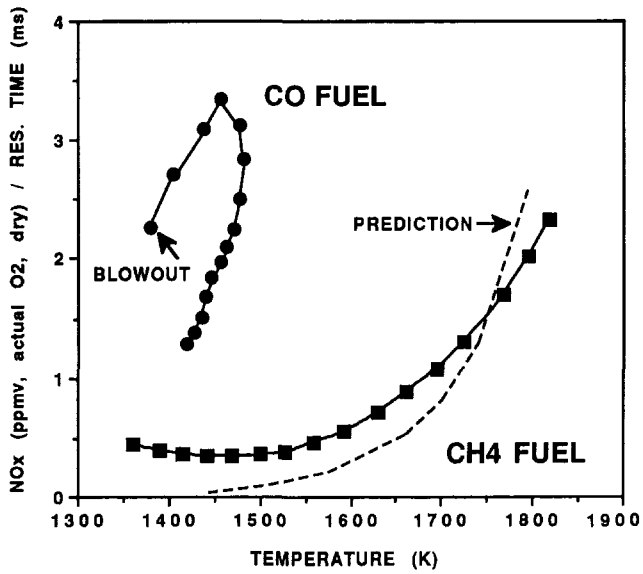

Fig. 1 Measured, overall rate of formation of NO_x, based on conditions of a dry sample at the actual O_2 concentration, versus temperature for a jet-stirred reactor operated at a pressure of 0.92 atm, an inlet temperature of 295 K, a fuel-air equivalence ratio of 0.75, and nominal mean residence times of 6 to 40 ms for the CH_4 fuel and 7 to 24 ms for the CO fuel, from [6]. Model predictions use the Miller and Bowman [2] chemical kinetic mechanism.

Table 1 Measured and predicted results for a jet-stirred reactor operated at 0.92 atm pressure and 295 K inlet temperature

FUEL	MEASUREMENTS			PREDICTIONS				
	T (K)	τ (ms)	NOx	Model	NOx	Prompt	Nitrous	Zeld.
CH4	1445	30.3	13.2	DB	1.51	1.23	0.24	0.04
				MB	1.03	0.31	0.68	0.04
				MWF	1.16	0.31	0.79	0.06
CH4	1661	11.5	9.4	DB	6.99	4.09	1.30	1.60
				MB	5.99	1.78	2.74	1.47
				MWF	7.27	1.78	3.56	1.93
CH4	1795	6.54	12.2	DB	18.13	6.52	2.74	8.87
				MB	16.79	3.86	4.82	8.11
				MWF	20.75	3.86	6.53	10.36
CO	1440	19.0	33.9	DB	5.20	0.02	4.80	0.33
				MB	18.19	0.0	17.90	0.29
				MWF	15.05	0.0	14.65	0.40
CO	1443	9.43	31.8	DB	3.95	0.02	3.67	0.26
				MB	13.54	0.0	13.30	0.24
				MWF	11.49	0.0	11.18	0.31

1 The importance of the nitrous oxide pathway to NO_x is demonstrated by the results for the combustion of the CO fuel. The nitrous oxide pathway contributed nearly all of NO_x. Furthermore, the rate of NO_x formation for the CO combustion was markedly greater than for the methane combustion (at similar temperatures). Because of the modest temperatures of the CO combustion, the Zeldovich pathway was weak. Model predictions indicate that less than 10 percent of the NO_x measured for the CO combustion was caused by the Zeldovich pathway (see Table 1). The CO combustion with the nearly dry air gave high O atom levels, which promoted N_2O formation by reaction (4 reverse), followed by N_2O oxidation to NO by reaction (5). The nitrous oxide was not destroyed by reaction (7), because of the low levels of the H atom. For the methane combustion, however, the O atom levels were lower, and the H atom levels were considerably greater. Thus, less N_2O was formed, and that which was formed was destroyed by reaction (7). Markedly less NO_x was formed.

An interesting additional aspect of the results for the CO combustion is the double-valued behavior of the NO_x. The lower part of the curve corresponds to the usual behavior of the reactor: i.e., the reactor temperature increased with decreasing mean residence time. The upper part of the curve, however, corresponds to the short mean residence times. The reactor was near blowout. The temperature (and free radical concentrations) increased with increasing mean residence time, resulting in the maximum in the NO_x formation rate at the temperature of about 1450K.

2 For the methane combustion, typical behavior is shown in Fig. 1 for temperatures greater than about 1600 K (corresponding to the shortest residence times run). That is, the rate of formation of the NO_x increased with increasing temperature, and the measured and modeled NO_x agree within a factor of about two. However, for temperatures below 1600 K, involving residence times of about 15 ms and greater, the reactor was significantly influenced by gradients caused by the finite mixing rate and the wall heat loss. The WSR is no longer a reasonable model. It significantly underpredicts the NO_x. It is interesting to note that at the lowest temperatures (and longest residence times) examined, the rate of NO_x formation measured actually increased with decreasing temperature. This is attributed to regions of elevated temperature in the center of the reactor. Because of the heat loss at the walls of the reactor, the temperatures in the center of the reactor, in the vicinity of the inlet jets, were hotter than the temperatures measured for the bulk of the reactor. This effect is explored in detail in [7].

Measured and modeled values for selected cases from Fig. 1 are given in Table 1. The predictions for the methane cases show that all the pathways contribute to the NO_x, except for a very weak Zeldovich contribution for the 1445 K case. As explained above, close agreement between the measured and modeled results is not expected for the methane cases because of the relatively long mean residence time run. On the other hand, the CO combustion more nearly satisfied the WSR condition, because of the modest temperatures and low H_2O concentrations. For these cases the lack of agreement between the measured and modeled results (using the MB and MWF mechanisms) is caused by uncertainty in the measured H_2O inlet air concentration. The CO combustion is very sensitive to changes in the H_2O content of nearly dry air. In [6], close agreement is obtained between the measured and modeled results for the NO_x when the measured O atom concentrations are used in a kinetic mechanism involving the nitrous oxide and Zeldovich pathways.

For the methane combustion, the DB, MB, and MWF mechanisms predict about the same total NO_x. The difference evident is the favoring by the DB mechanism of prompt NO_x over NO_x formed by the nitrous oxide pathway. For the CO combustion, the reduced prediction by the DB mechanism of the NO_x formed by the nitrous oxide pathway is significant. The MB and MWF mechanisms appear to offer the best representation of the nitrous oxide pathway.

Comparison to the Model

Figure 1 and Table 1 are useful primarily for the trends they show. The measurements considered now show very good agreement to the model predictions, and thus, are useful quantitatively. These are the results shown in Fig. 2.

In Fig. 2, results of Corr et al. [7], Bartok et al. [8], Malte et al. [9], and Singh et al. [10] are plotted. The fuel-rich data of [8] and [9] have not been included, and only single data points have been used from [7] and [10], since the remainder of their data were obtained in combustion gases diluted by an inert in order to maintain a constant temperature. Also, not all of the lean data of [8] are plotted, in order to show a relatively uncluttered figure.

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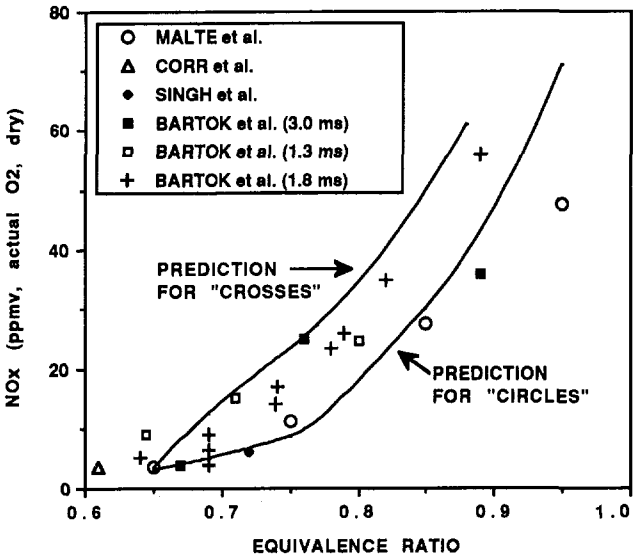

Fig. 2 Measured NO_x (dry, actual O_2) versus fuel-air equivalence ratio for jet-stirred reactors operated at ambient pressure and nominal mean residence times of 1.3 to 5 ms. Bartok et al [8] measurements for 464 K inlet temperature; other measurements (from [7, 9 and 10]) for 295 K. Model predictions use the Miller and Bowman [2] chemical kinetic mechanism.

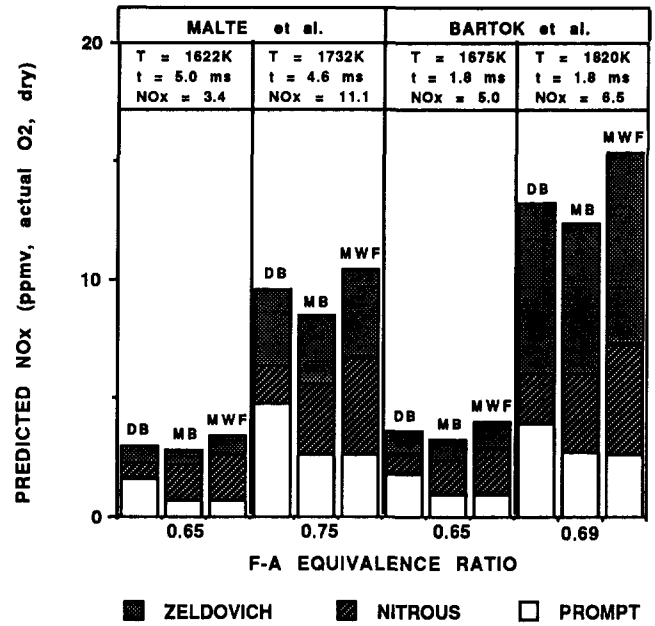

Fig. 3 Predicted contributions of the Zeldovich, nitrous oxide, and prompt chemical pathways to total NO_x using the Drake and Blint (DB), the Miller and Bowman (MB), and the Michaud et al. (MWF) kinetic mechanisms, and assuming a well-stirred reactor operating at the measured temperature and mean residence time, versus fuel-air equivalence ratio. Same conditions as for Fig. 2.

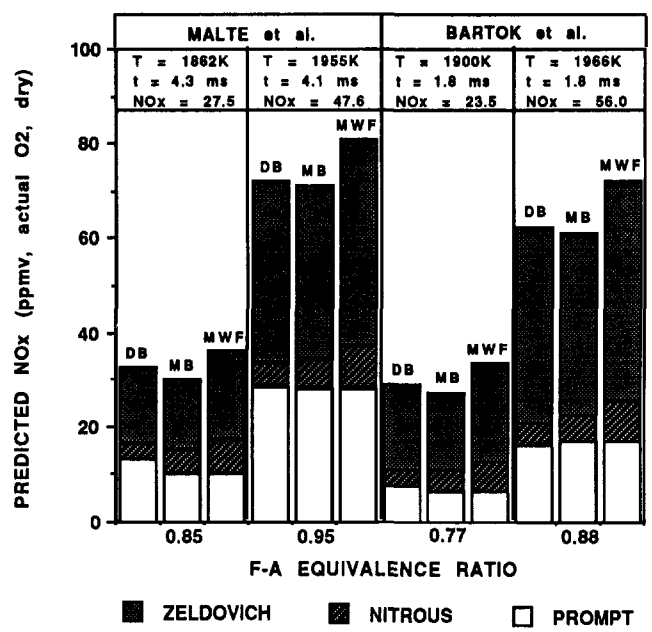

Fig. 4 Predicted NO_x contributions assuming a well-stirred reactor operating at the measured temperature and mean residence time versus fuel-air equivalence ratio. Similar to Fig. 3.

The range of ϕ is 0.60 to 0.95, and the range of τ is 1.3 to 5.0 ms. The pressure is 1 atm for [7] and [8], and 0.92 atm for [9] and [10]. The reactant inlet temperature is 464 K for [8], and 295 K (nominal) for the other data.

Because the temperatures of Malte et al [9] were measured by OH absorption and also by fine, coated, heat loss corrected thermocouples, these temperature data are assumed to be the most accurate of the datasets.

Figure 2 shows NO_x predictions using the MB mechanism, applied to the experimental conditions of Bartok et al. [8] at 1.8 ms mean residence time, and Malte et al. [9]. These cases have been selected for modeling because of the extensive database of [8] at 1.8 ms (and the use of [8] for kinetic mechanism development [5]), and the accurate temperatures of [9]. (In [4], comparison of all the measurements shown in Fig. 2 to the model with the DB mechanism was conducted.)

In Fig. 2 the agreement between the measured and predicted NO_x values is very good, especially for the conditions of Malte et al. [9] for $\phi < 0.90$. For $\phi > 0.90$, the rate of combustion was sufficiently rapid for the free radical concentrations (O, OH, and H) to relax toward equilibrium in the bulk of the reactor volume. Thus, as confirmed by direct measurement of OH, the free radical concentrations in the bulk of the reactor were less than the WSR-predicted values [9]. This, in turn, meant a decrease in the amount of NO_x formed by the Zeldovich and nitrous oxide pathways. This is an example of the influence of finite mixing causing a reduction in NO_x compared to the WSR condition. (This is in contrast to the behavior shown in Fig. 1 and discussed above, in which case the slow mixing led to increased NO_x compared to the WSR prediction.)

With respect to the results of Bartok et al. [8], two features are noted: (1) The NO_x is greater than that measured by Malte et al. [9] because of the higher inlet temperature. Also, the very short residence times of [8] led to enhanced free radial concentrations, and therefore, to greater NO_x yields. (2) The small differences noted between the modeled and measured values are assumed to be due to uncertainty in the measured temperature.

The generally good agreement noted in Fig. 2 between the modeled and measured NO_x is taken as verification of the model for lean-premixed combustion, at least for well-stirred combustion at atmospheric pressure.

NO_x Pathways. Attention is now turned to the contributions of the different NO_x pathways for the cases of Fig. 2 modeled. These results are shown in Figs. 3 and 4. The experimental conditions for the cases of [8] and [9] treated are listed in the boxes near the tops of the figures. The following characteristics are found:

1 Especially for the results shown in Fig. 3, with ϕ between

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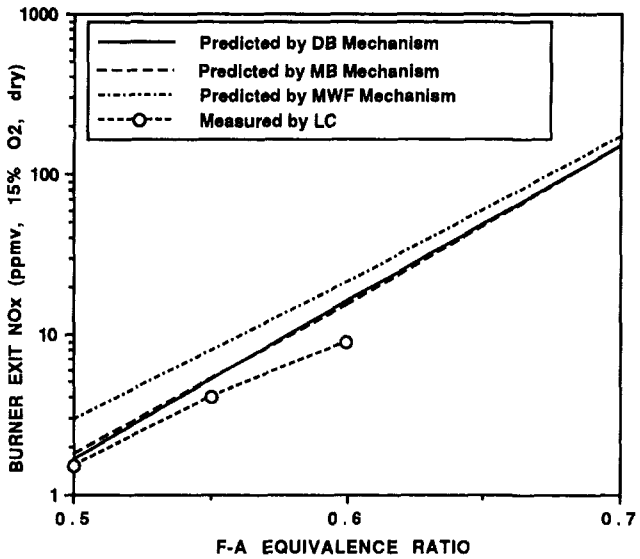

Fig. 5 Predicted and measured NO_x for a porous-plate burner operated at a pressure of 10.4 atm, and an inlet temperature of 616 K versus fuel-air equivalence ratio. Measurements are from Leonard and Correa [11]. Predictions are based on an adiabatic chemical reactor consisting of a WSR operating at incipient blowout, followed by a PFR.

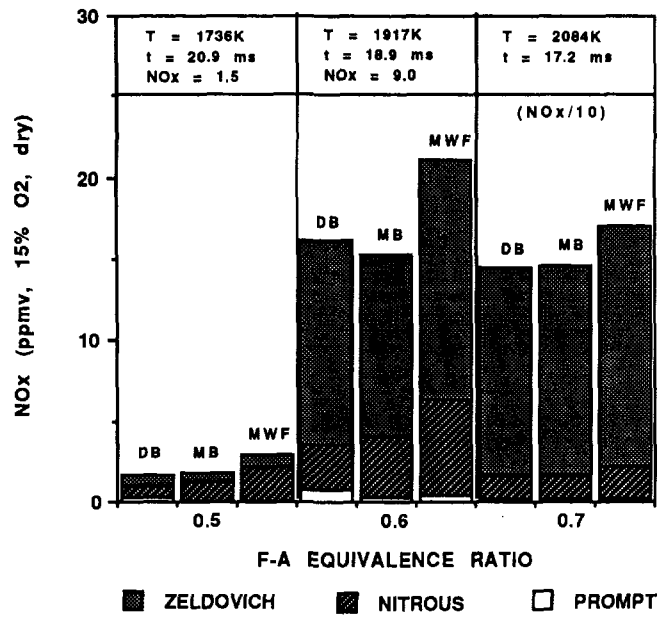

Fig. 6 Predicted NO_x contributions for the porous-plate burner versus fuel-air equivalence ratio. Same conditions as for Fig. 5. The NO_x results for the 0.7 equivalence ratio are divided by 10.

0.65 and 0.75, all three NO_x pathways are significant. None is negligible.

2 The total NO_x predicted is about the same for the different mechanisms, though, because of the inclusion of the singlet O atom chemistry, the MWF mechanism generally gives a slightly greater NO_x than the DB and MB mechanisms.

3 At the leanest (and lowest temperature) cases considered, the nitrous oxide and prompt pathways provide the majority of the NO_x. Furthermore, the mechanisms differ. For the DB mechanism, the greatest contributor is the prompt pathway; whereas for the MB and MWF mechanisms, the greatest contributor is the nitrous oxide pathway. (It should be noted, that the prompt NO_x portion of the MWF mechanism is based on the MB mechanism.)

From these results it is concluded that all three NO_x pathways must be considered for lean, premixed combustion, at least for well-stirred combustion at atmospheric pressure.

Porous-Plate Burner NO_x

Attention is now directed to the porous-plate burner operated at elevated pressure. The questions are (1) whether the model and chemical kinetic mechanisms are valid for this burner, and (2) whether the NO_x formation is similar to that for the jet-stirred reactor.

Experimental Conditions. The porous-plate burner data are from Leonard and Correa [11], hereafter termed measurements by LC. The fuel was methane, premixed with air. The burner pressure was 10.4 atm, and the mixture inlet temperature was 616 K. (Leonard and Correa also conducted experiments for lower pressures and mixture inlet temperatures. Those results were modeled previously by the present authors [4].) The length of the burner, i.e., the distance from the porous-plate to the sampling point, is 11.4 cm, and the burner diameter is 5 cm. This database considered covers a range of ϕ from 0.5 to 0.6.

Comparison to the Model. The burner is modeled by applying the baseline model (i.e., the WSR operating at incipient blowout, followed by the PFR). Adiabatic behavior is assumed.

In Fig. 5, the NO_x modeled by the three mechanisms is compared to the measurements. Results are shown for the experimental ϕ range of 0.5 to 0.6, as well as for assumed operation at ϕ up to 0.7. Over the ϕ range from 0.5 to 0.7, by the adiabatic assumption, the variation in the overall burner τ is 20.4 to 17.2 ms, and the variation in the burner exit temperature is 1736 to 2084 K (by all three mechanisms). The corresponding values predicted for the WSR zone operating at incipient blowout are $\tau = 0.25$ to 0.02 ms and $T =$ 1643 and 1795 K (DB mechanism), and $\tau = 0.21$ to 0.03 ms and $T = 1611$ to 1860 K (by the MB and the MWF mechanisms).

For the leanest and coolest case considered ($\phi = 0.5$), good agreement is noted between the measured NO_x and the NO_x predicted by the model using the DB and MB mechanisms. Differences between the modeled and measured NO_x for the higher ϕ examined are attributed to heat loss from the burner, which is not considered in the model.

Michaud et al. [3] developed their mechanism through comparison to porous-plate burner data very similar to that examined here. In their study [3], the burner heat loss was accounted for, and good agreement between predicted and experimental data was claimed. For the present application, assuming adiabaticity, the predictions based on the MWF mechanism are a few parts per million greater than those obtained based on the DB and MB mechanisms.

With respect to the first question posed above, it appears, at least from the chemical kinetic standpoint, that the model is valid. The model agreement to the laboratory measurements is reasonable, and can be improved by computing the burner heat loss.

NO_x Pathways

In Fig. 6, the breakdown of the NO_x contributions is shown. These results are for the exit of the burner. Listed in the boxes near the top of the figure are the predicted adiabatic exit conditions, and the measured NO_x.

With respect to the prompt NO_x, the behavior is considerably different from that of the jet-stirred reactor cases stud-

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ied above. At the leanest and coolest conditions examined the nitrous oxide pathway is predominant. The prompt NO_x is small to negligible. The Zeldovich NO_x is secondary. However, for ϕ greater than about 0.6, the Zeldovich NO_x is predominant. (The inclusion of heat loss in the model would shift the regime of Zeldovich predominance to slightly higher ϕ).

Thus, the second question posed above is also answered. The NO_x behavior of the porous-plate burner operated at elevated pressure is different from that of the jet-stirred reactors operated at ambient pressure. For lean-premixed combustion at low NO_x levels, the nitrous oxide pathway is predicted to be predominant in the porous-plate burner, whereas for the jet-stirred reactors, the predictions indicate that both the nitrous oxide and prompt pathways are important. There are at least three effects involved in this regard:

1 The elevated pressures promote the nitrous oxide pathway, because of the three-body recombination reaction (4 reverse), which initiates the pathway.

2 The elevated inlet temperatures (of the porous plate burner) diminish the prompt pathway.

3 Hydrocarbon free radicals (such as CH, which enters into the prompt NO_x initiating reaction (12)) are restricted to a narrow zone in the porous-plate burner. They are not dispersed throughout the burner, as is possible in the jet-stirred reactor operated at very intense mixing rates.

Comments on the Residence Time Distribution

The particle residence time distributions of the two types of laboratory reactors examined are much different, and cover the limiting possibilities. For the jet-stirred reactors, the individual particles have residence times from very short to very long compared to the mean residence time; whereas, for the porous-plate burner, all of the particles reside in the burner for approximately the mean residence time. It is noteworthy, as demonstrated above, that these distinctly different situations both exhibit good agreement between the measured and modeled NO_x. This provides confidence in the model and the chemical kinetics.

The lean-premixed primary zone of gas turbine engine combustors should have a particle residence time distribution that lies between the limiting distributions of the laboratory reactors. The tendency, however, may be toward a relatively narrow residence time distribution (centered around the mean residence time). This distribution is provided by the WSR/PFR baseline model (described above and used to simulate the porous-plate burner). In order to examine the effect a wider residence distribution has on the NO_x, the fraction of the combustion process treated as a WSR is enlarged (as described above). Although there is virtue in identifying a chemical reactor model that best simulates the actual combustor, the objective of this study regarding the importance of the nitrous oxide pathway is well met by applying the simple WSR/PFR model, and by varying the WSR to PFR ratio.

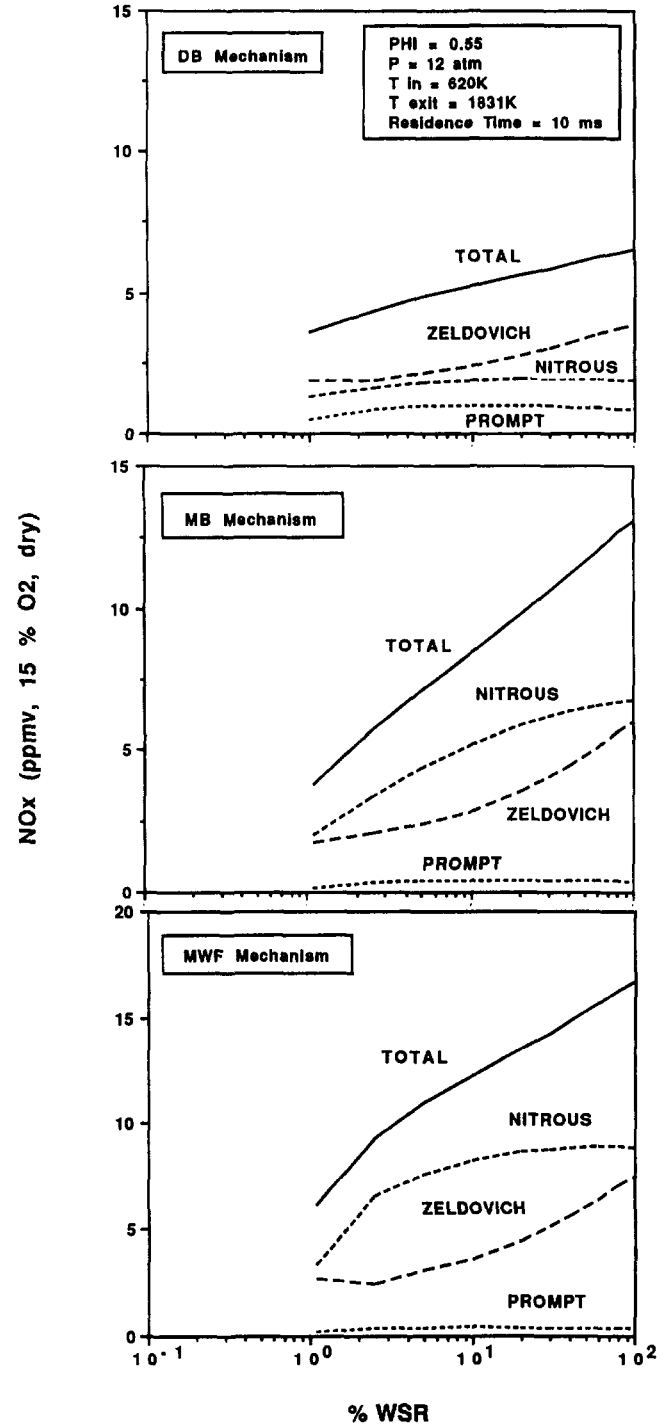

Fig. 7 Predicted NO_x contributions versus percentage of the combustor modeled as a WSR. Balance of combustor is a PFR. Total mean residence time of the combustor is fixed at 10 ms. Combustor pressure of 12 atm, inlet temperature of 620 K, and a fuel-air equivalence ratio of 0.55.

Lean-Premixed Combustion Modeled Results

The results modeled for the NO_x formed in the lean-premixed primary zone of the gas turbine combustor are now presented and discussed.

Effect of WSR Fraction. A decision in the type of modeling conducted here is the size (or fraction of the total primary zone combustor volume) chosen for the WSR. The remainder of the combustor is the PFR. The WSR is the region of elevated free radical concentrations. The baseline model assumes a WSR of size corresponding to incipient blowout. Elsewhere [3], the WSR size has been selected to be a constant value, corresponding to the size for blowout of the leanest mixture that can be burned at the P and T_{in} treated.

NO_x predictions for lean-premixed combustion are sensitive to the size of the WSR. This is shown in Figs 7, 8, and 9, for three assumed gas turbine combustor conditions. The results in Figs. 7 and 8 are for an intermediate pressure combustor (12 atm and 620 K combustor inlet temperature) having a primary zone ϕ of 0.55 and 0.65, respectively. Note

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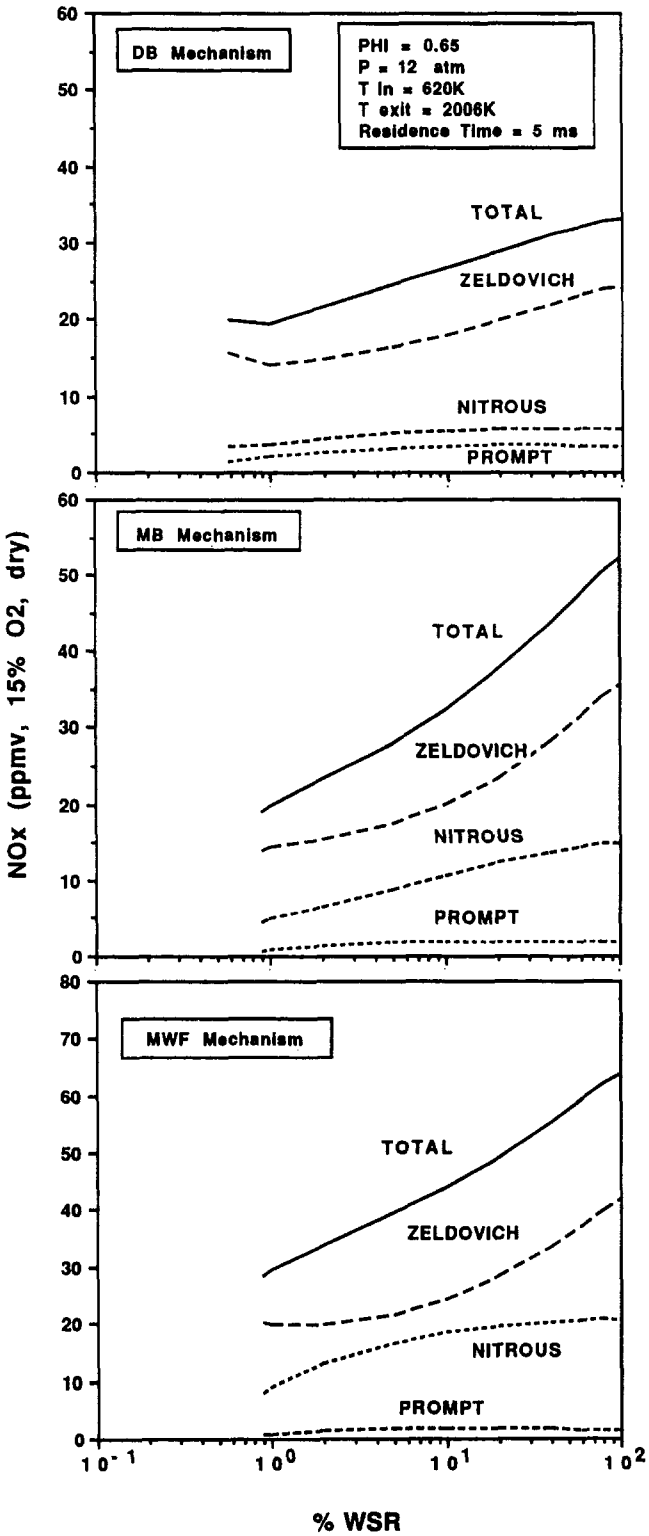

Fig. 8 Predicted NO_x contributions versus percentage of combustor modeled as a WSR. Balance of combustor is a PFR. Total mean residence time of the combustor is fixed at 5 ms. Combustor pressure of 12 atm, inlet temperature of 620 K, and a fuel–air equivalence ratio of 0.65.

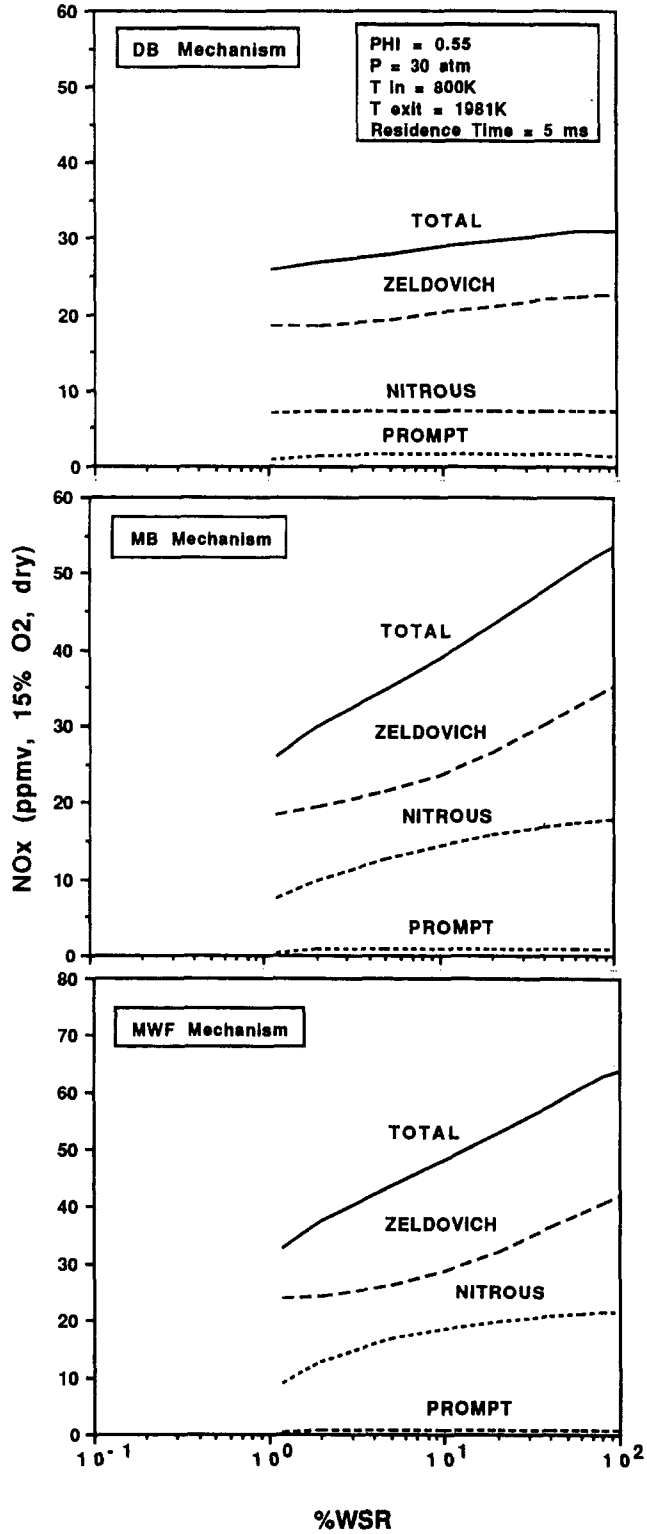

Fig. 9 Predicted NO_x contributions versus percentage of combustor modeled as a WSR. Balance of combustor is a PFR. Total mean residence time of the combustor is fixed at 5 ms. Combustor pressure of 30 atm, inlet temperature of 800 K, and a fuel air equivalence ratio of 0.55.

that the mean residence time of the $\phi = 0.55$ combustor has been increased to $\tau = 10$ ms (from the nominal value of 5 ms used herein, in order to overcome the lower temperature of this case and effect reasonable completion of combustion). In Fig. 9, the results are shown for a high pressure combustor (30 atm and 800 K combustor inlet temperature) operated at $\phi = 0.55$ and $\tau = 5$ ms.

For all of the cases treated, the percentage of the primary zone volume modeled as the WSR is increased from the blowout value (about 1 percent) to 100 percent. The major trends are summarized as follows:

1 In all cases the NO_x increases with increasing WSR size. The MB mechanism (followed closely by the MWF mech-

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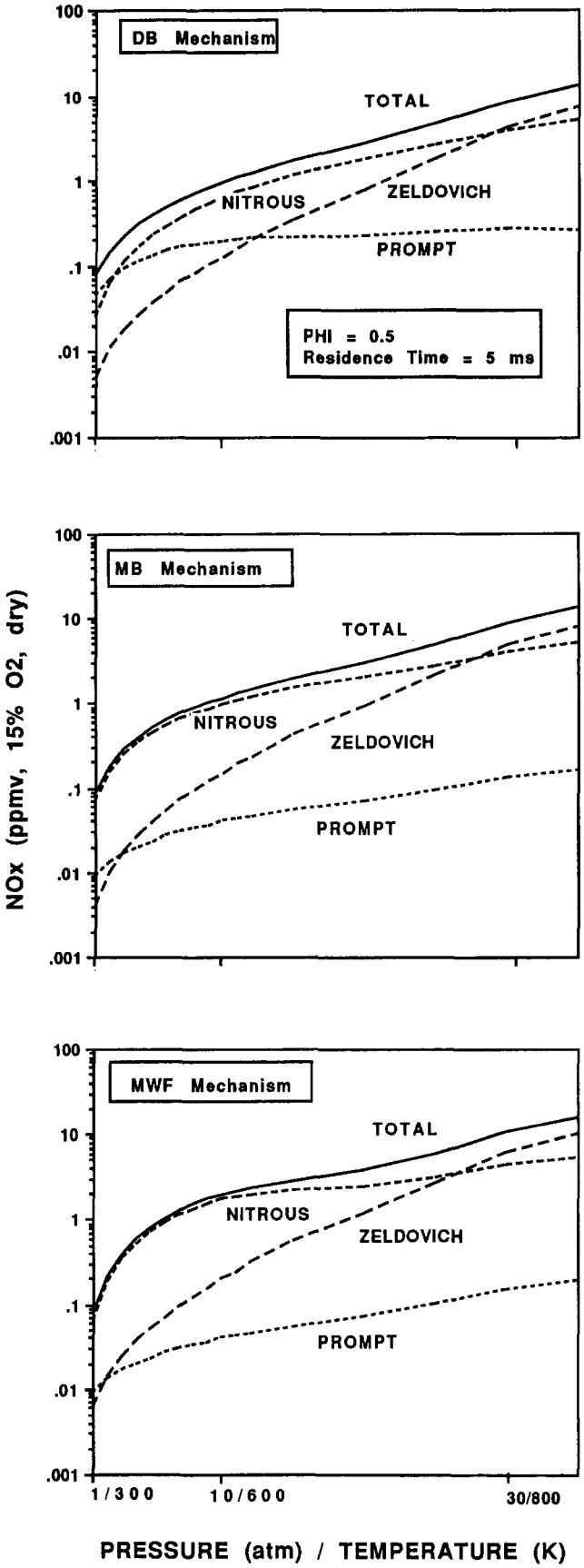

Fig. 10 Pressure and inlet temperature effects on NO_x predicted for a lean-premixed combustor modeled as a WSR operating at incipient blowout, followed by a PFR. The combustor inlet temperature is calculated assuming an engine compressor of 85 percent isentropic efficiency drawing standard air. The mean residence time is 5 ms and the fuel-air equivalence ratio is 0.5.

anism) shows the greatest sensitivity. The DB mechanism for the 30 atm combustor shows the least sensitivity.

2 The Zeldovich contribution shows the greatest sensitivity to the WSR percentage. The nitrous oxide contribution tends to level off above about 10 percent WSR percentage.

3 The MWF mechanism predicts the largest NO_x amounts.

4 The order of the NO_x contribution is Zeldovich > nitrous oxide > prompt. The exception is the 12 atm, $\phi = 0.55$ case, for which the order is nitrous oxide > Zeldovich > prompt, when the MB and MWF mechanisms are used.

5 The contribution of the prompt pathway is small to negligible. This is true even for the DB mechanism, which gives the highest prompt NO_x contribution. Even for the case of 100 percent WSR, the prompt NO_x is never as strong as the nitrous oxide contribution. This is different from the behavior noted in Fig. 3 for the atmospheric pressure jet-stirred reactors operated at $\phi = 0.65$, for which case the prompt NO_x predicted by the DB mechanism was predominant.

These results imply that the lean-premixed flame should be thin in order to effect minimum NO_x formation. That is, the flame should be designed to permit the free radicals to relax toward equilibrium as quickly as possible, after ensuring that the essential hydrocarbon oxidation steps are completed. A drawn-out zone of free radical chemistry, as obtained with large-scale recirculation, would appear to enhance the NO_x. (This situation is different from that of diffusion flames, for which long, slowly mixed flames have been found to reduce NO_x.)

Effects of Pressure and Inlet Temperature. The remainder of the results are for the baseline model (i.e., WSR at incipient blowout, and total residence time of 5 ms). The combined effect of pressure and combustor inlet temperature on NO_x is shown in Fig. 10 for $\phi = 0.5$. The pressure range is 1 to 35 atm. Pressure increments in regions of significant sensitivity to pressure are 1 atm; elsewhere the increments are 5 atm. The corresponding combustor inlet temperatures are approximately those at the exit of a gas turbine engine compressor of 85 percent isentropic efficiency inducting standard sea level air.

The pressure sensitivity of the NO_x (without the influence of the combustor inlet temperature) may be assessed by assuming that NO_x varies as pressure raised to the exponent n according to the equation:

$$NO_x(\text{ppmv})\,\alpha P^n \tag{17}$$

Results for the exponent n are listed in Table 2 for the range of pressures and inlet temperatures considered herein. Conditions for which the inlet temperature is matched to the pressure by the engine compressor assumption used above are indicated by the bold typeface. The values of n are derived from logarithmic straight line fits to the computed results around 1, 10, and 30 atm. The traditional pressure exponent (for thermal NO_x and for Fenimore [12] prompt NO_x) is 0.5.

The results of Fig. 10 and Table 2 are summarized as follows:

1 As shown in Fig. 10, the greatest sensitivity of the NO_x to pressure (and inlet temperature) occurs in the range of approximately 1 to 8 atm pressure. The weakest sensitivity occurs in the range of approximately 10 to 13 atm. This range of weakest sensitivity is noted especially for the results obtained with the MWF mechanism.

2 In the 1 to 10 atm range, most of the NO_x sensitivity noted in Fig. 10 is caused by the response of the nitrous

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Table 2 Pressure dependence of NO_x for lean-premixed combustion at a mean residence time of 5 ms

	DB Mechanism				MB Mechanism				MWF Mechanism			
	T\P	1	10	30	T\P	1	10	30	T\P	1	10	30
Φ=0.5	300	**0.42**	--	--	300	**0.50**	--	--	300	**0.50**	--	--
	600	0.13	**0.67**	1.13	600	0.26	**0.19**	0.00	600	0.21	**-0.13**	-0.44
	800	0.13	0.19	**0.62**	800	0.22	0.20	**0.30**	800	0.19	0.06	**0.14**
	T\P	1	10	30	T\P	1	10	30	T\P	1	10	30
Φ=0.6	300	**0.26**	0.80	0.70	300	**0.33**	0.31	-0.15	300	**0.28**	-0.15	-0.66
	600	0.18	**0.27**	0.57	600	0.26	**0.22**	0.28	600	0.23	**0.09**	0.14
	800	0.33	0.35	**0.52**	800	0.37	0.38	**0.48**	800	0.35	0.33	**0.44**
	T\P	1	10	30	T\P	1	10	30	T\P	1	10	30
Φ=0.7	300	**0.24**	0.45	0.80	300	**0.31**	0.24	0.08	300	**0.27**	0.00	-0.20
	600	0.40	**0.37**	0.51	600	0.44	**0.38**	0.47	600	0.41	**0.33**	0.42
	800	0.61	0.48	**0.51**	800	0.62	0.49	**0.50**	800	0.58	0.43	**0.44**

oxide pathway to changing pressure (and inlet temperature). Above approximately 10 atm, the response of the nitrous oxide pathway is relatively weak. This is noted especially for the MWF mechanism, in which case the response of the NO_x formed by the nitrous oxide pathway to increasing pressure (and inlet temperature) is almost flat above approximately 10 atm. For the highest pressures examined, the NO_x sensitivity is influenced by the response of the Zeldovich pathway to pressure (and inlet temperature, especially).

3 The results in Table 2 show that there is no universal pressure exponent for the NO_x formed in lean-premixed combustion.

4 The results in Table 2 show reasonable agreement between the DB, MB, and MWF mechanisms when the following conditions are true: $P = 1$ atm (all inlet temperatures and ϕ's considered); $P = 10$ atm and $T_{in} = 600$ K (all ϕ's considered except 0.5); $P = 10$ atm and $T_{in} = 800$ K (all ϕ's considered): $P = 30$ atm and $T_{in} = 600$ K ($\phi = 0.7$ only); $P = 30$ atm and $T_{in} = 800$ K (all ϕ's considered except 0.5). For the leanest ϕ's considered (i.e., 0.5) the agreement between mechanisms is poor, except when the pressure is low or the inlet temperature is high.

5 For a lean-premixed combustor operating at $P = 10$ atm, $T_{in} = 600$ K, and $\phi = 0.6$, the results in Table 2 indicate that the NO_x pressure exponent lies in the 0.1 to 0.3 range. As P, T_{in}, or ϕ increase, the exponent increases (toward approximately 0.5). On the other hand, as P, T_{in}, or ϕ decrease, the exponent becomes quite dependent on the mechanism selected, and no clear behavior is evident.

6 For a lean-premixed combustor operating at $P = 30$ atm, $T_{in} = 800$ K, and ϕ of either 0.6 or 0.7, the results in Table 2 indicate that the NO_x pressure exponent is approximately 0.5. For a leaner value of ϕ, the exponent becomes very dependent on the mechanism selected.

7 Although generally in this study the NO_x concentration predicted does not depend strongly on the mechanism selected, this tendency does not carry over to the pressure dependence of the NO_x for gas turbine engine conditions when ϕ is about 0.5 (and less). For the very lean fuel–air ratios, more study will be required in order to understand the pressure dependence of the NO_x.

The Nitrous Oxide Contribution

In Fig. 11, the percentage contribution of the nitrous oxide pathway to the total NO_x is plotted versus the absolute amount of the NO_x formed by this pathway. For gas turbine engine, lean-premixed combustor conditions, the minimum contribution is about 10 percent. A significant percentage contribution is noted when the absolute contribution to the NO_x is in the 1 to 10 ppmv (15 percent O_2 dry) range.

In Fig. 12, the percentage contribution of the nitrous oxide pathway is plotted versus the total NO_x. These results indicate that whenever the total NO_x emission is in the 15 to 30 ppmv (15 percent O_2, dry) range, the nitrous oxide pathway contributes 40 to 45 percent of the NO_x for a high pressure engine (30 atm) and 20 to 35 percent of the NO_x for an intermediate pressure engine (10 atm). For engines producing less than 10 ppmv (15 percent O_2, dry) total NO_x, the nitrous oxide contribution increases steeply and approaches 100 percent at the lowest total NO_x levels.

Conclusions

In this paper, total NO_x and the contributions of the nitrous oxide, Zeldovich, and prompt pathways in lean-premixed combustion have been presented. The kinetic mechanisms of Drake and Blint [1], Miller and Bowman [2], and Michaud et al. [3] have been compared over a range of operating conditions. The computer model has been compared to laboratory data. The modeled results for NO_x showed good agreement to the laboratory data for jet-stirred reactors operated at short mean residence times, and for the porous-plate burner operated at minimum heat loss conditions.

A comparison of the relative importances of the nitrous oxide, Zeldovich, and prompt pathways for gas turbine engine conditions shows that the three kinetic mechanisms considered give similar results for most of the conditions examined. All three mechanisms predict comparable total NO_x amounts. The DB mechanism consistently predicts the largest prompt NO_x contribution. The MB and the MWF mechanisms, which use identical hydrocarbon chemistry, pre-

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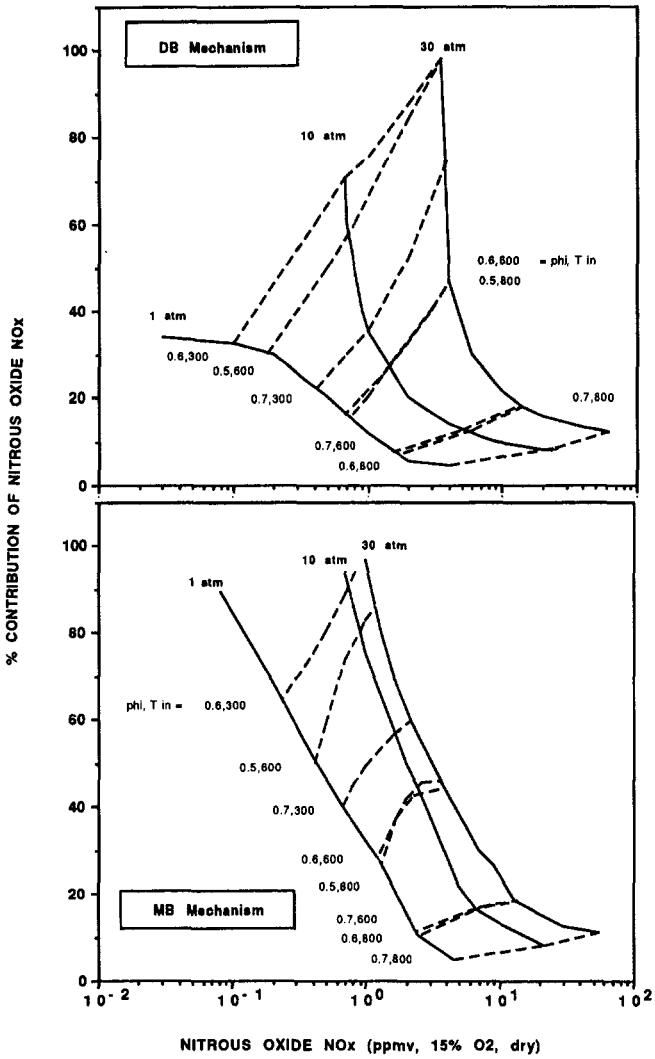

Fig. 11 Predicted percent contribution of nitrous oxide-NO_x versus absolute nitrous oxide-NO_x for a lean-premixed combustor modeled as a WSR operated at incipient blowout, followed by a PFR. Mean residence time equal to 5 ms.

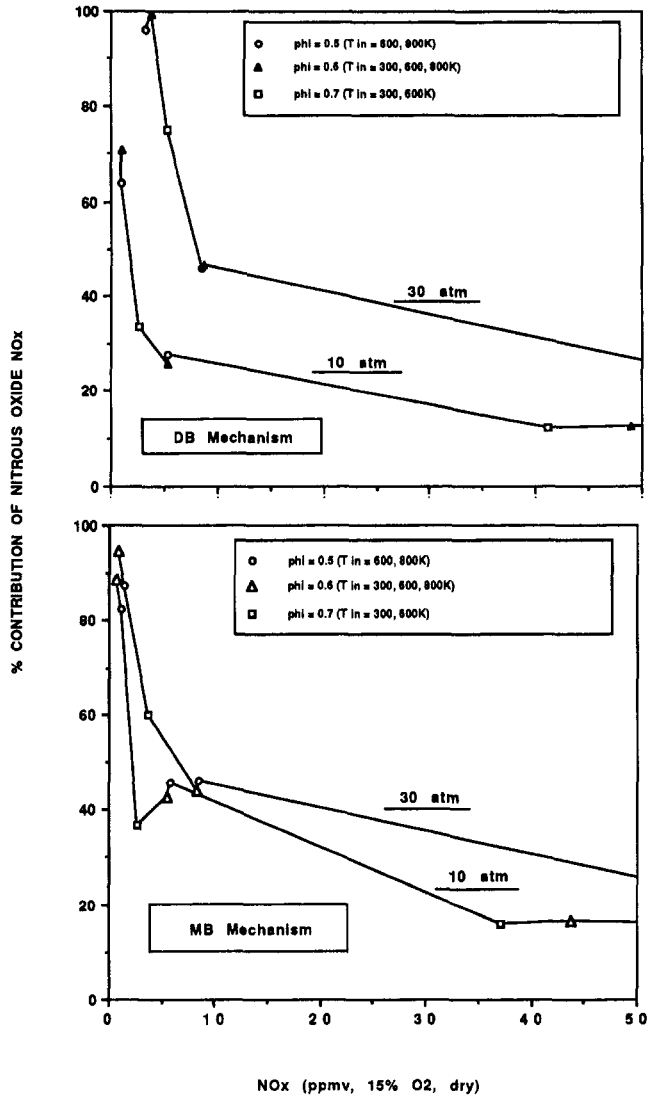

Fig. 12 Predicted percentage contribution of nitrous oxide-NO_x versus total NO_x for a combustor modeled as a WSR operated at incipient blowout, followed by a PFR. Mean residence time equal to 5 ms.

dict the same prompt NO_x contributions. The MWF mechanism consistently predicts greater nitrous oxide and Zeldovich NO_x, and hence, greatest total NO_x, because of the inclusion of the singlet O atom chemistry. The two areas of significant difference between the three mechanisms involve (1) the pressure dependence of the NO_x for the leanest cases examined (i.e., $\phi = 0.5$), and (2) the low nitrous oxide–NO_x predicted by the DB mechanism for the CO combustion experiments. These areas should receive further study.

Over the range of pressures, combustor inlet temperatures, and residence times considered, for the leanest conditions examined, the major source of the NO_x exiting the combustor is formed by the nitrous oxide pathway. At richer conditions, all three kinetic mechanisms show that the largest NO_x source is the Zeldovich pathway, but that the nitrous oxide pathway is an nonnegligible route.

The results indicate that whenever the total NO_x emission is in the 15 to 30 ppmv (15 percent O_2, dry) range, the nitrous oxide pathway contributes 40 to 45 percent of the NO_x for a high-pressure engine (30 atm) and 20 to 35 percent of the NO_x for an intermediate pressure engine (10 atm). For engines producing less than 10 ppmv (15 percent O_2, dry) total NO_x, as is the goal with ultra low NO_x combustors, the nitrous oxide contribution increases steeply and approaches 100 percent.

Although this study has examined NO_x formation from the chemical kinetic viewpoint, the results indicate that the fluid mechanics of the lean-premixed combustor should be arranged so that the flame zone(s) are as thin as possible. Broad zones of high-intensity, recirculating combustion are likely to carry elevated free radical concentrations, and thus, increase the NO_x formation.

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